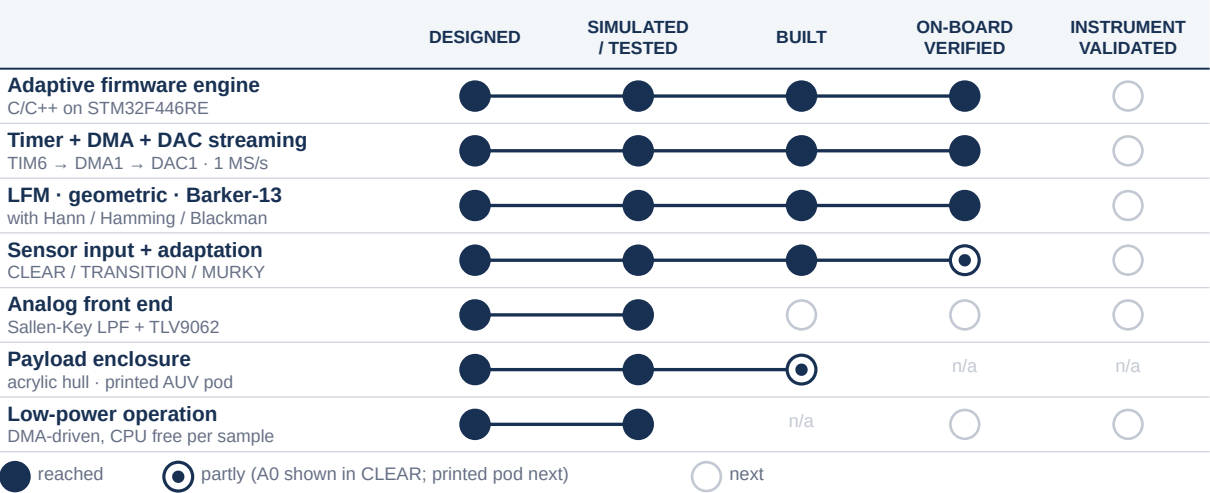
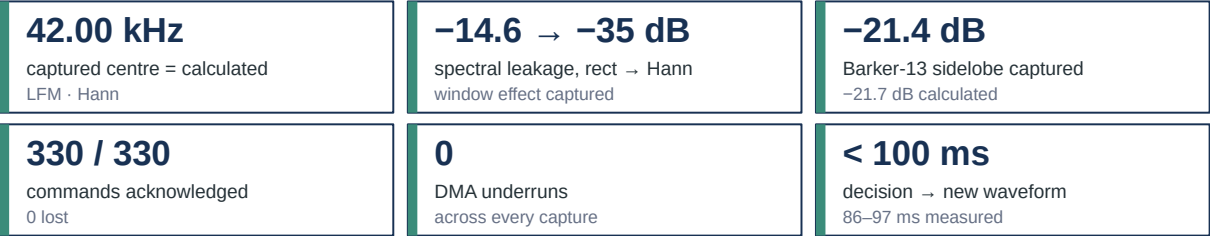


ANALYSIS OF FEASIBILITY

Readiness of every subsystem, from design to instrument validation



PROVEN ON THE BOARD · MCU ADC capture and firmware telemetry, not oscilloscope data



VIABILITY	Low-cost COTS STM32 Nucleo · TLV9062 · standard passives	Software-defined new waveform or policy = firmware update	Scalable same architecture → 100–500 kHz with an external DAC	Maintainable 161 automated tests guard every change
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CHALLENGES & RISKS → STRATEGIES

Each risk paired with how we overcome it

CHALLENGE	STRATEGY	STATUS
Frequency scale PS cites 100–500 kHz; bench transducer is 40 kHz	→ Frequency-agnostic policy table; external high-speed DAC on the same TIM + DMA path	SCALE PATH
No oscilloscope on hand external validation not yet possible	→ On-board ADC + DMA capture with FFT and matched filter now; lab scope at judging, CSV import ready	IN PLACE
Analog stage not yet built filter + op-amp still to assemble	→ Two independent models agree within 0.02 dB in band; stock-value parts list ready	NEXT BUILD
Noisy or drifting sensor input a bad reading could change the ping	→ Smoothing, hysteresis, stale and out-of-range rejection; TDS labelled as a proxy	IN PLACE
CPU load and battery budget per-sample trig math is expensive	→ Next pulse computed between pings, DMA streams it; inline-meter power test planned	IN PLACE
Loss of control link ping must never run unattended	→ Output off at boot, 2 s control lease auto-stop, DAC parked mid-scale on any fault	IN PLACE
Water ingress payload must survive the hull slot	→ O-ring end caps; parametric pod compiled and fit-checked in OpenSCAD	DESIGNED